

L7

Mid-Semester Examination

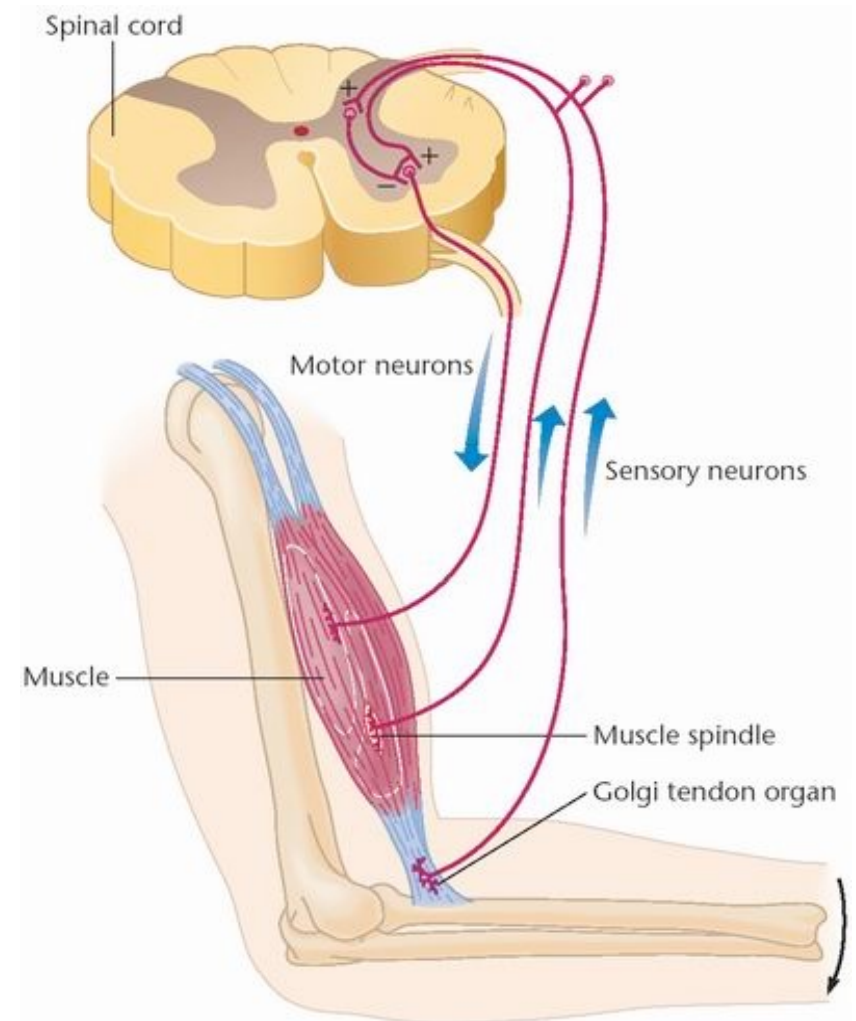
- 25th September 3-4 pm
 - Three sections
 - Total -30 marks
 - Section A: 15 questions → **15 marks**
 - Quiz Format
 - Attempt any 12
 - Section B -- > **8 marks**
 - Differences
 - Short answers
 - Definition
 - Section C -- > **10 marks**
 - Long answers

Proprioception

Perception of information from receptors present in the muscles, joints, tendons and skin.

Kinesthesia

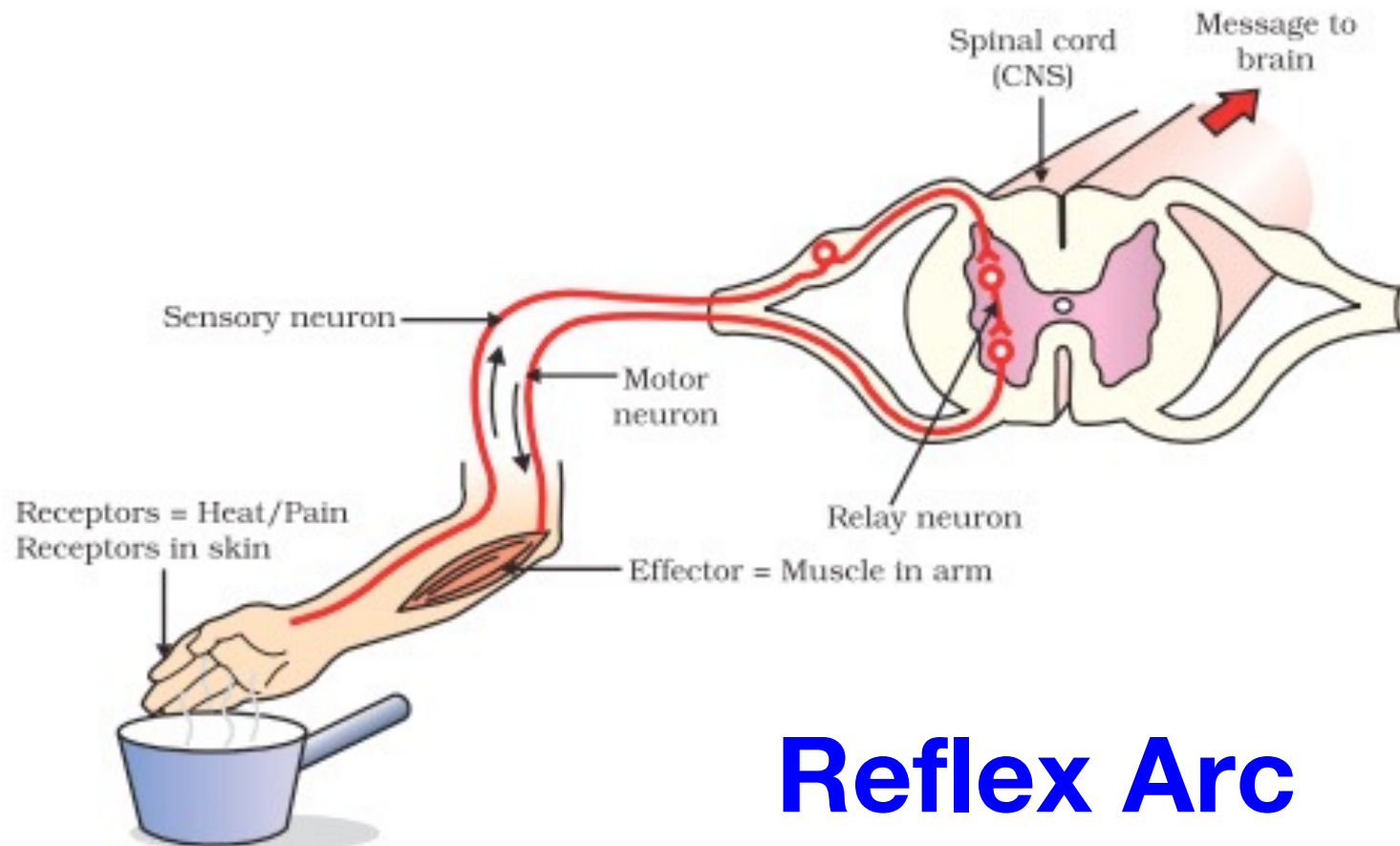
- Muscle spindles
- Golgi tendon organs
- Joint receptors
- Cutaneous receptors



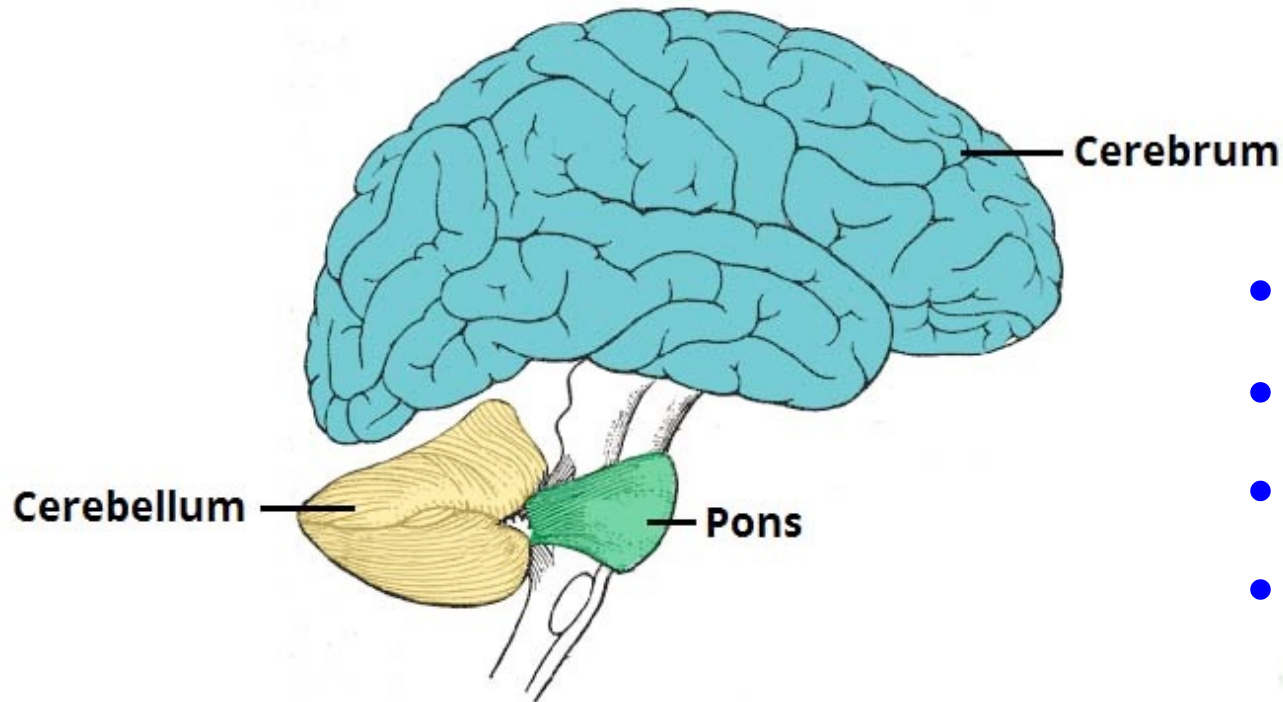
Spinal Cord

- Relay center
 - Recurrent connections
 - Reciprocal connections
- Reflex action

Reflex Action



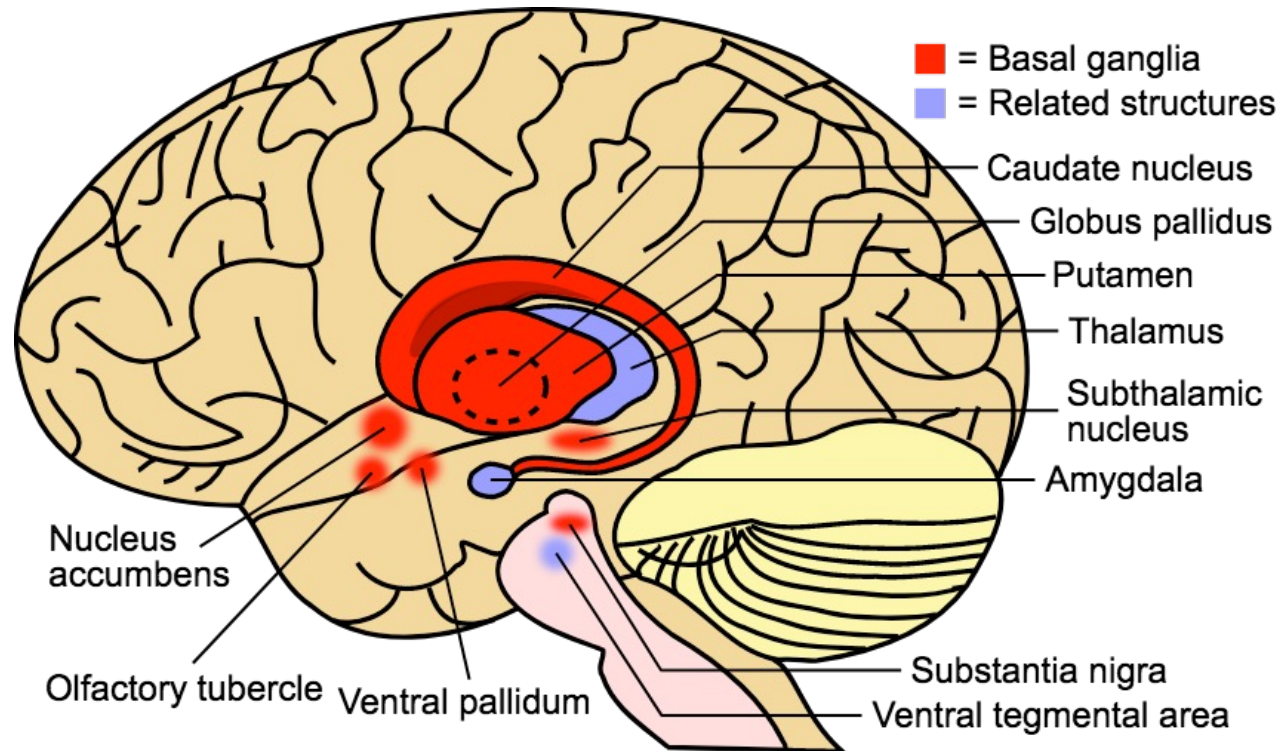
Cerebellum



- **Muscle Tone**
- **Coordination**
- **Timing**
- **Learning**

<https://teachmeanatomy.info/neuroanatomy/structures/cerebellum/>

Basal ganglia



https://en.wikipedia.org/wiki/Basal_ganglia

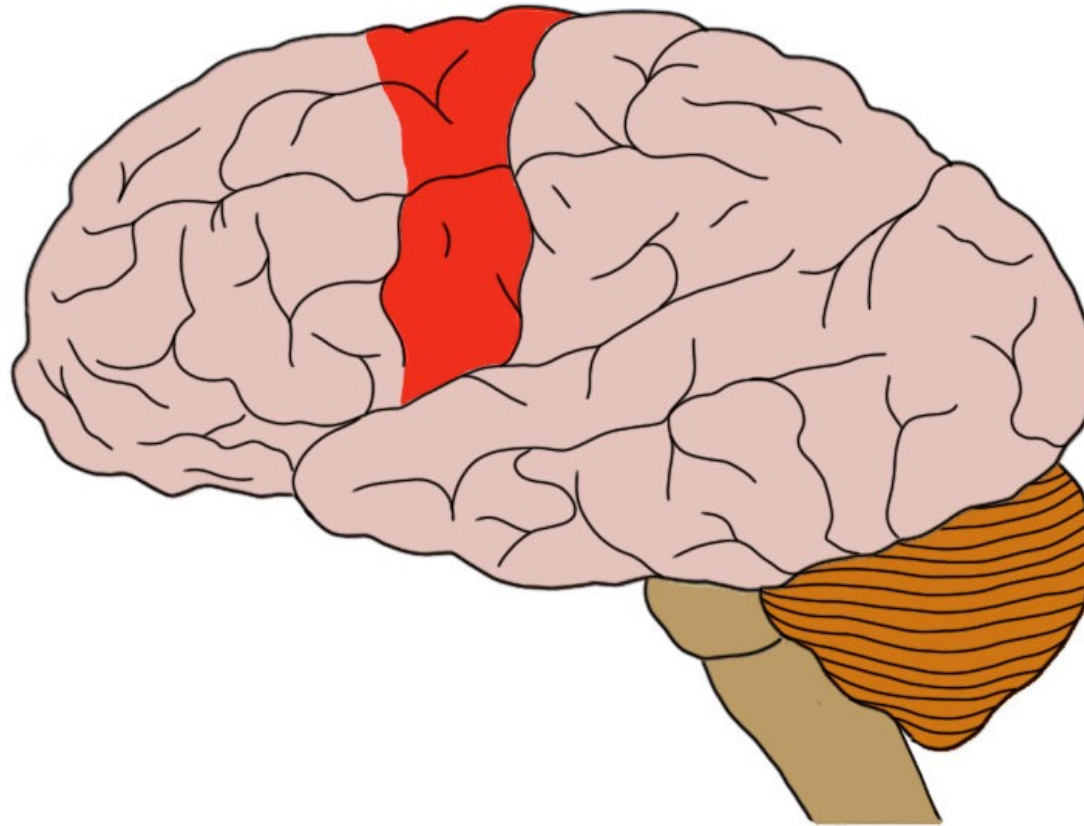
- Retrieval of movement plans
- Scaling amplitudes of movement
- Perceptual motor integration

Homunculus



<https://www.amazon.co.uk/The-Original-Homunculus-Company-Sensory/dp/B076YBYHMN>

Motor Cortex



<https://www.neuroscientificallychallenged.com/blog/know-your-brain-motor-cortex>

ON THE RELATIONS BETWEEN THE DIRECTION OF TWO-DIMENSIONAL ARM MOVEMENTS AND CELL DISCHARGE IN PRIMATE MOTOR CORTEX

APOSTOLOS P. GEORGOPOULOS, JOHN F. KALASKA, ROBERTO CAMINITI AND JOE T. MASSEY
1981

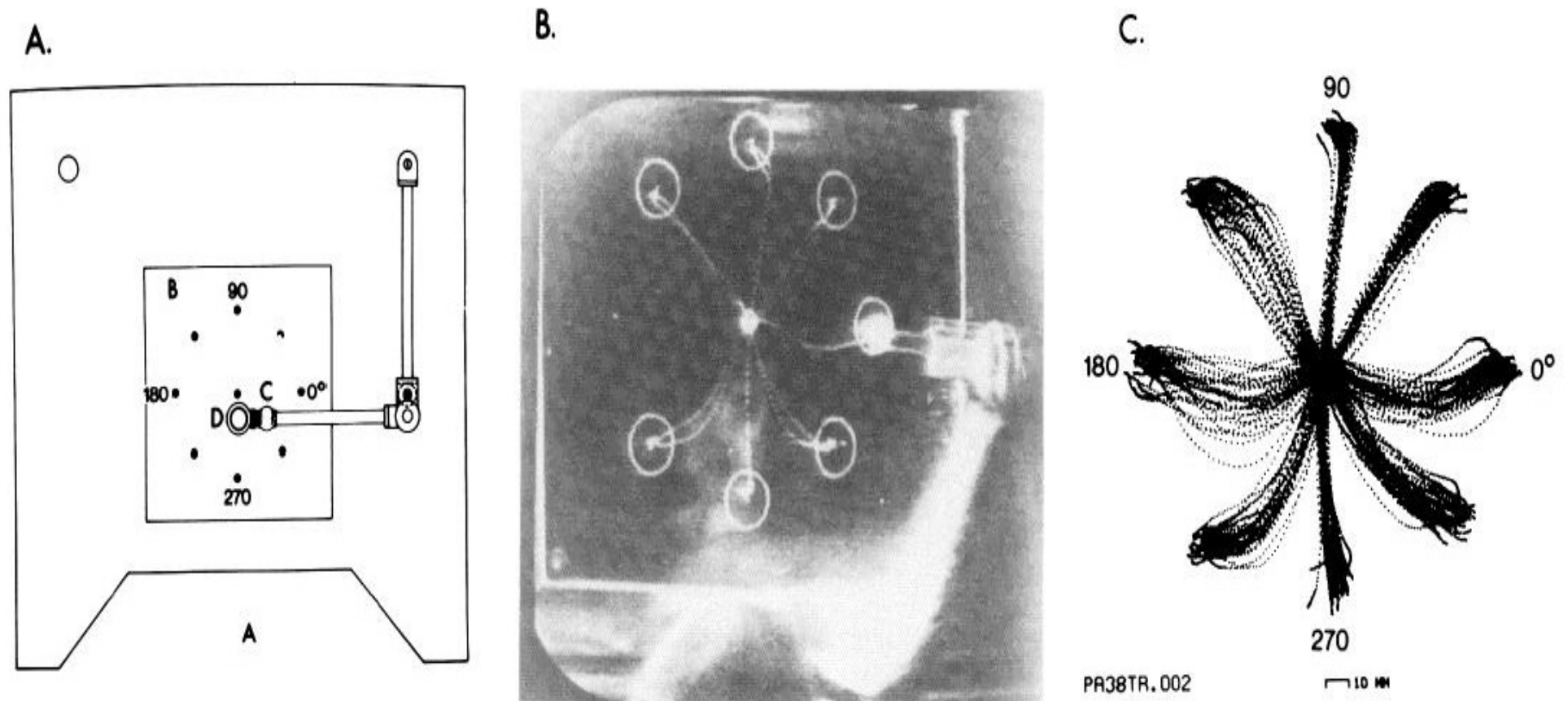
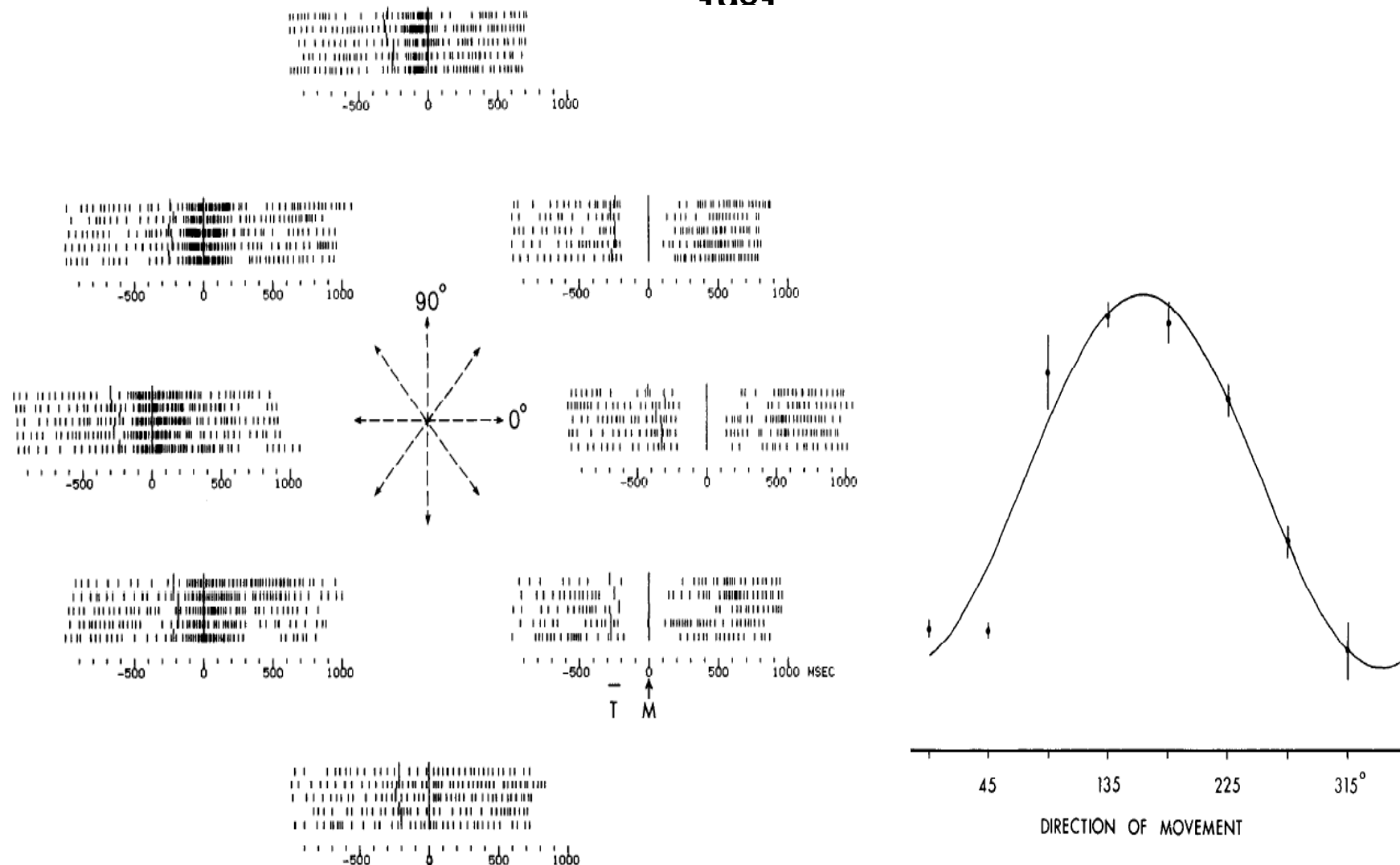


Figure 1. A. Diagram of the behavioral apparatus. B. Micrograph of a motor cortex area. C. Polar plot of arm movement directions.

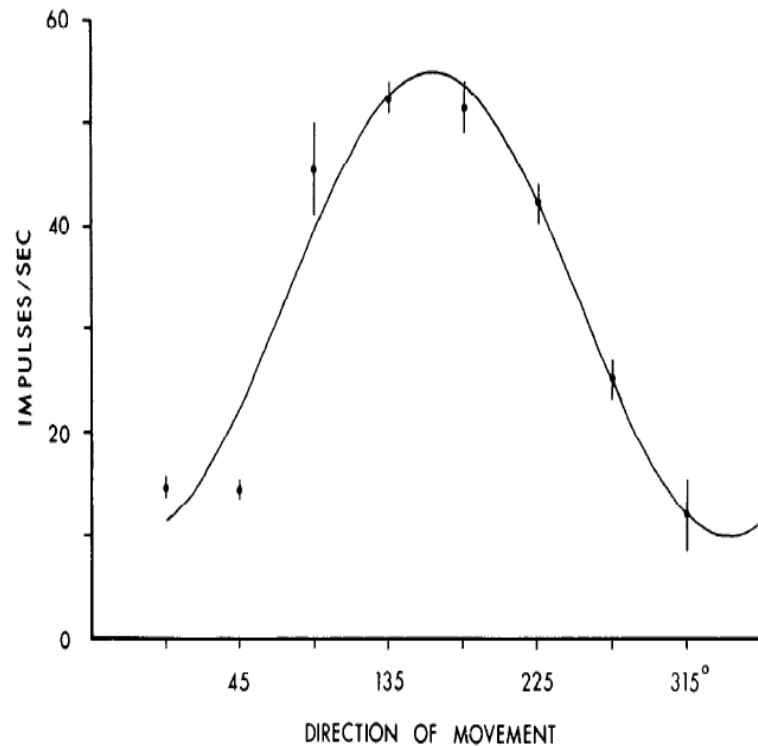
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**ON THE RELATIONS BETWEEN THE DIRECTION OF TWO-DIMENSIONAL ARM
MOVEMENTS AND CELL DISCHARGE IN PRIMATE MOTOR CORTEX**
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1981



- **Direction tuning**
- **Population coding**

Muscle Movement

Population Coding

Direction	Strength of Firing
90	100%
90 $\pm$ 1 89/ 91	99%
90 $\pm$ 2 88/ 92	98%
90 $\pm$ 3 87/ 93	97%
90 $\pm$ 4 86/94	96%
90 $\pm$ 5 85/95	95%

Direction	Strength of Firing
80	100%
80 $\pm$ 1 79/ 81	99%
80 $\pm$ 2 78/ 82	98%
80 $\pm$ 3 77/ 83	97%
80 $\pm$ 4 76/84	96%
80 $\pm$ 5 75/85	95%
80 $\pm$ 10 70/90	90

**For 85 degrees =
95 + 95 = 190**

Population Coding

Direction	Strength of Firing
90	100%
90 ± 1 89/ 91	99%
90 ± 2 88/ 92	98%
90 ± 3 87/ 93	97%
90 ± 4 86/94	96%
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80 ± 5 75/85	95%
80±10 70/90	90

**For 90 degrees =
95 + 95 = 190**

Population Coding

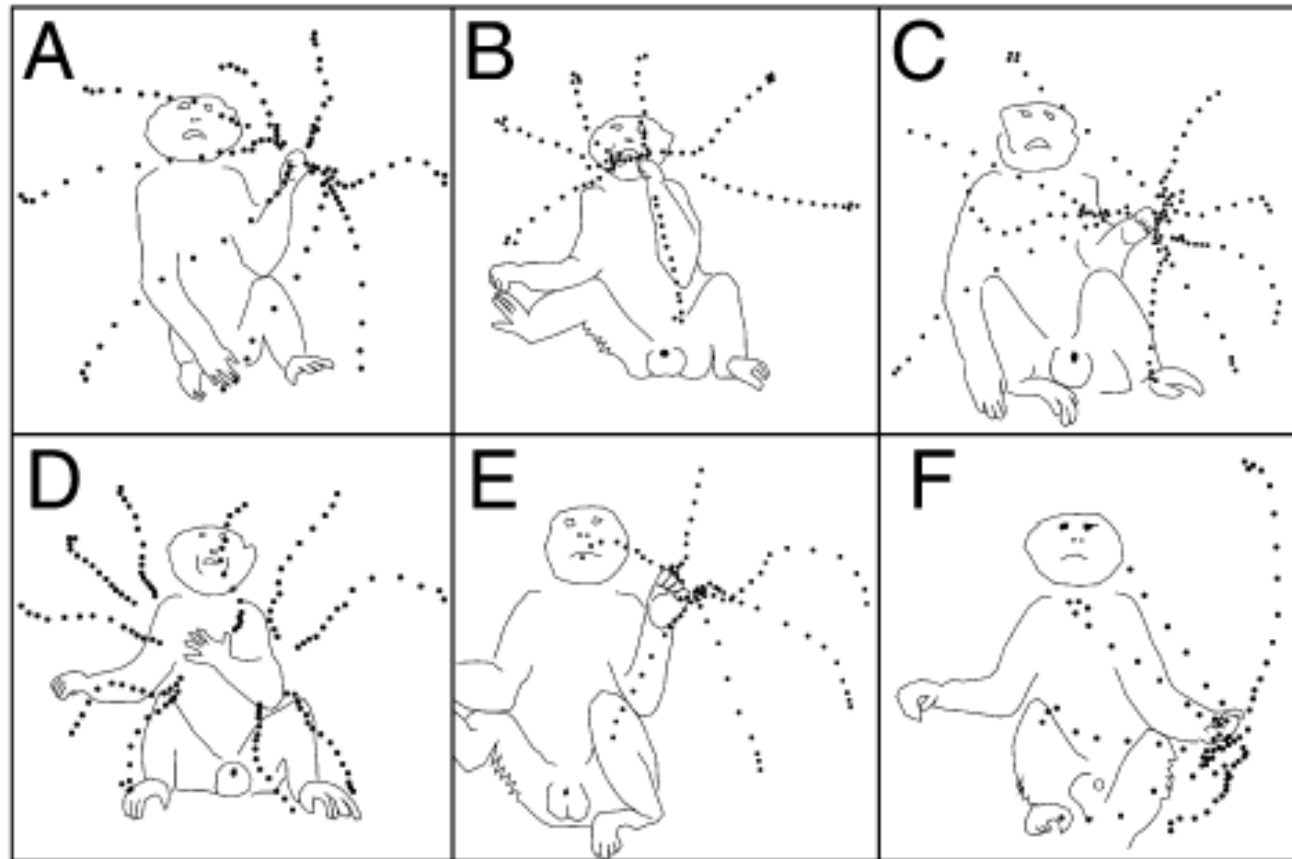
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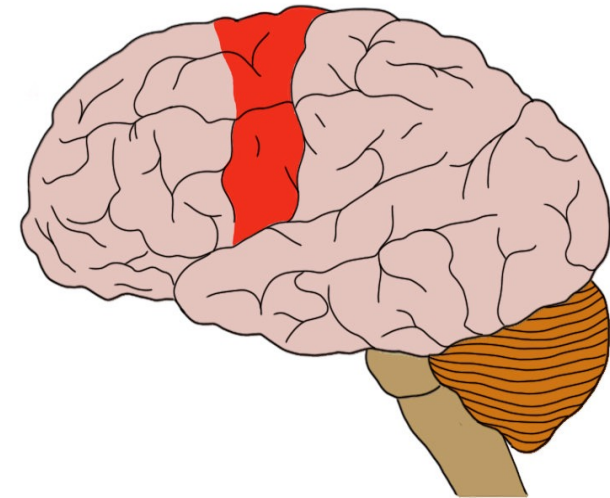
Complex Movements Evoked by Microstimulation of Precentral Cortex
Michael S.A. Graziano,¹Charlotte S.R. Taylor, and Tirin Moore
2002



Microstimulation initiated various kind of postures

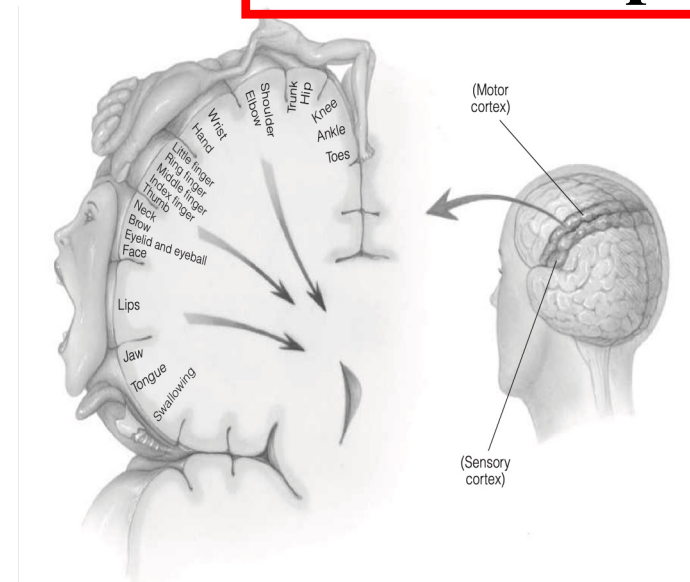
Motor Cortex

- Topographic representation
- Muscle movement
- Postures
- Some neurons
 - Amplitude
 - Speed

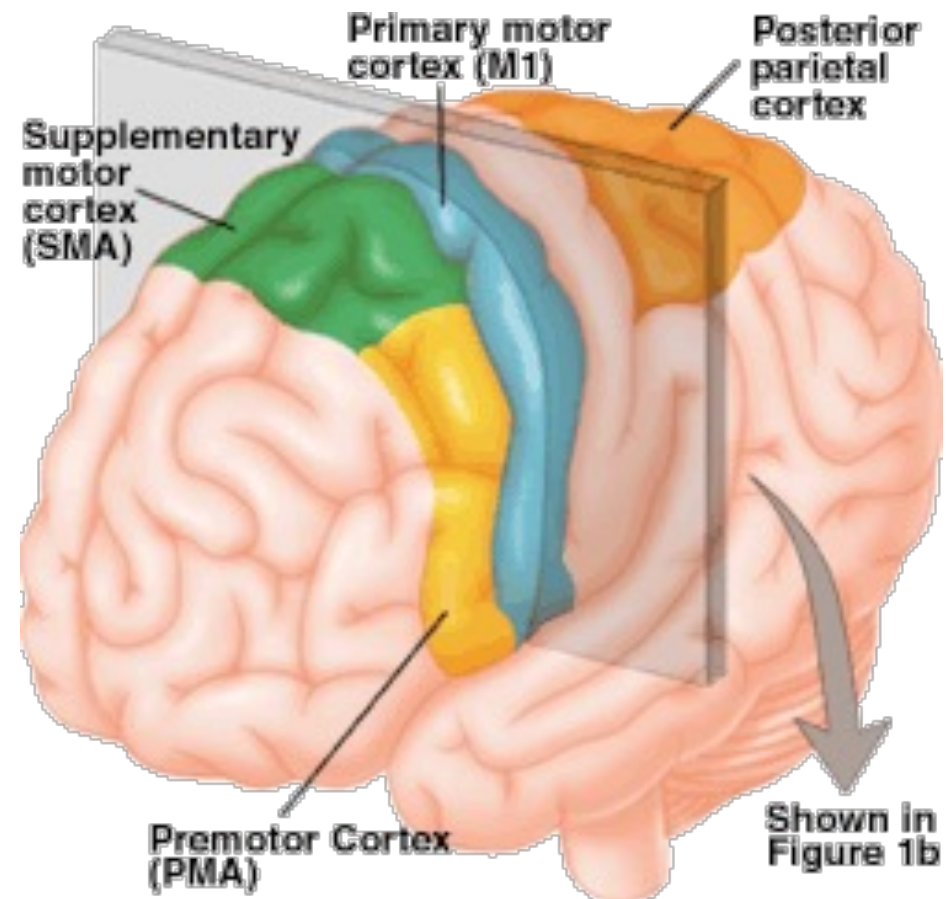


<https://www.neuroscientificallychallenged.com/blog/know-your-brain-motor-cortex>

Motor Map



Premotor Cortex



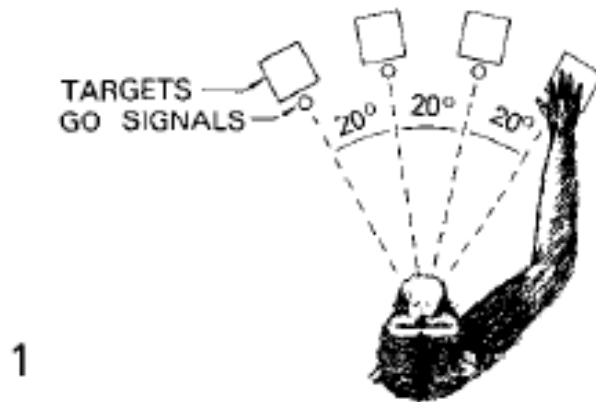
<https://brainconnection.brainhq.com/2013/03/05/the-anatomy-of-movement/>

THE PREMOTOR CORTEX OF THE MONKEY

MICHAEL WEINRICH AND STEVEN P. WISE

1982

INTER-TRIAL:



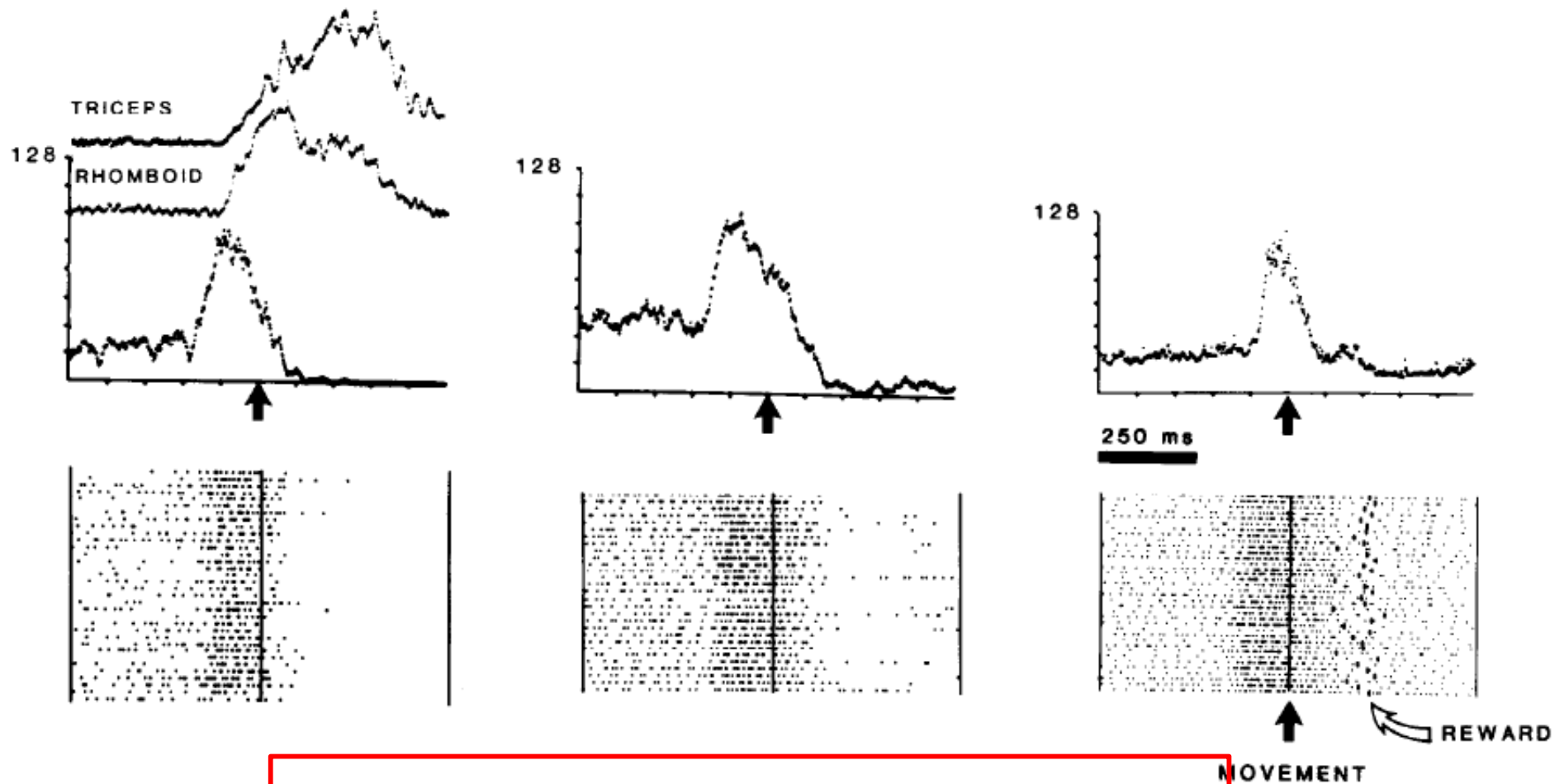
READY:



THE PREMOTOR CORTEX OF THE MONKEY

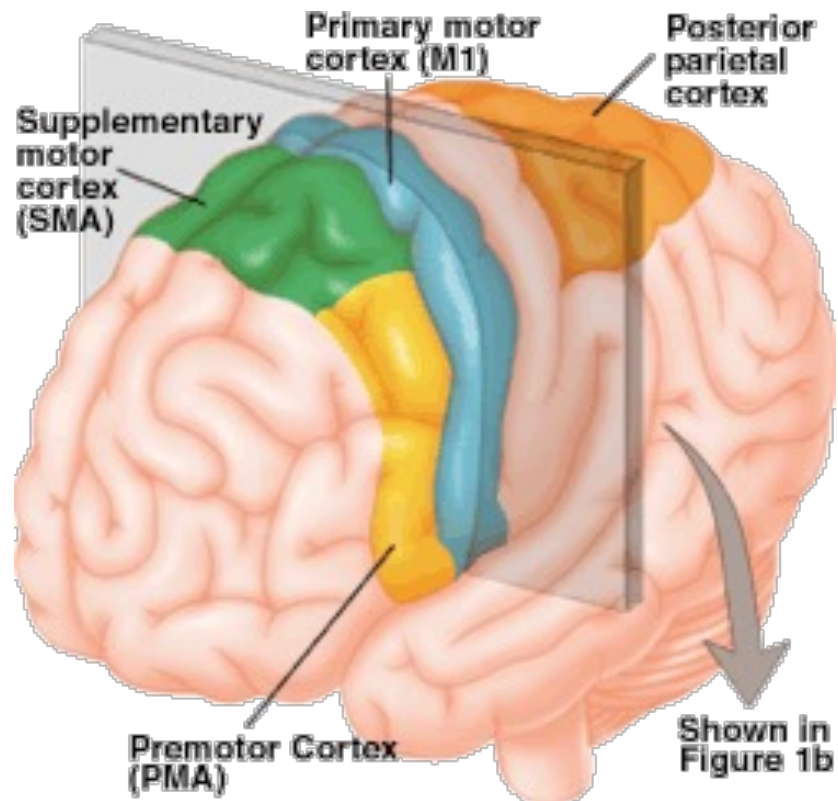
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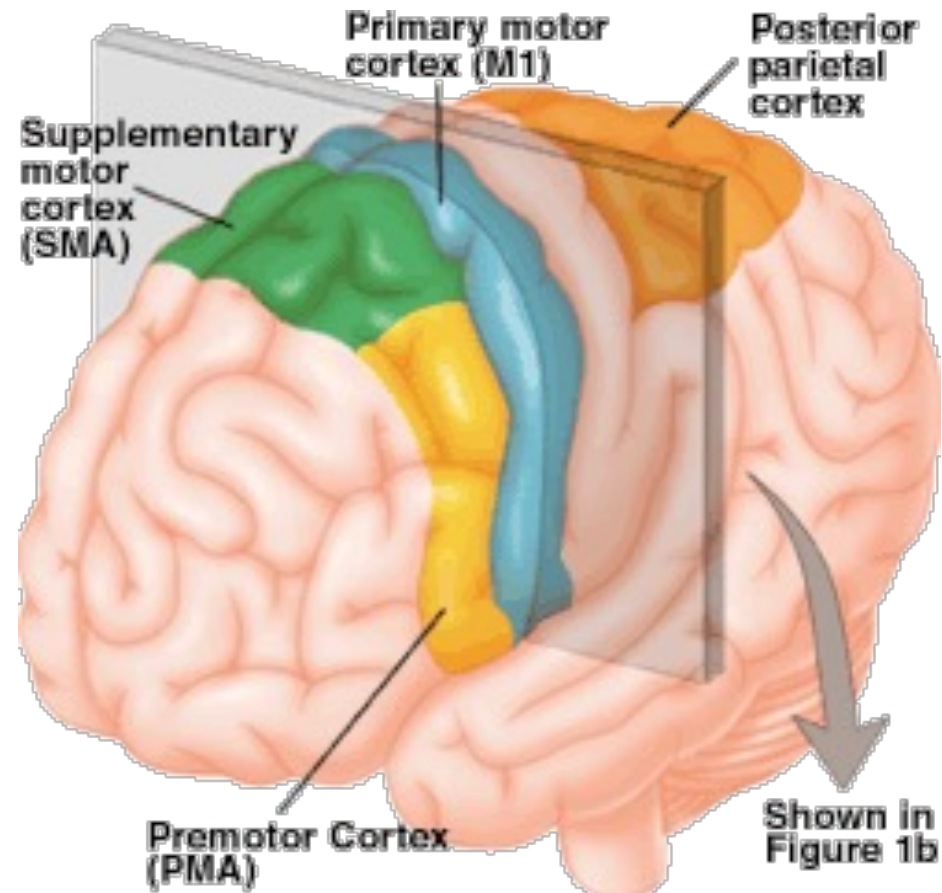
Movement preparation

Premotor Cortex



- **Movement preparation**
- **Trajectory selection**
- **Cross modal sensory integration**

Supplementary Motor cortex



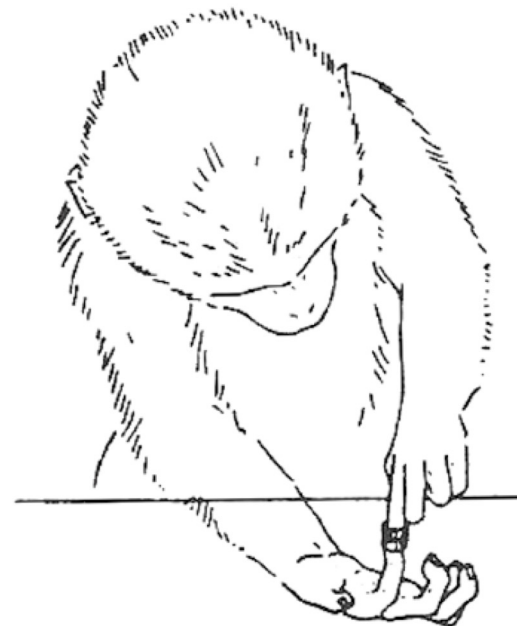
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Movement planning

Lesions in supplementary motor area interfere with a monkey's performance of a bimanual coordination task
Cobie Brinkman
1981



Normal animal



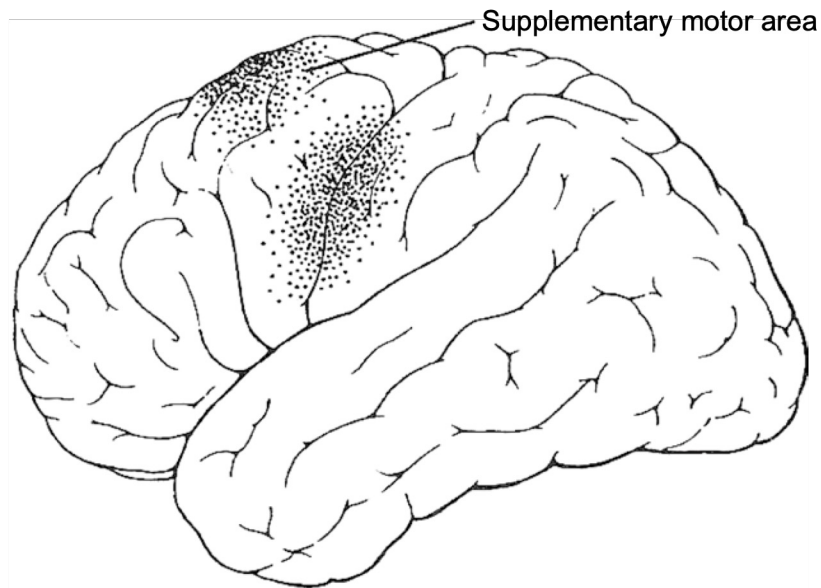
5 months after right SMA lesion

Movement planning

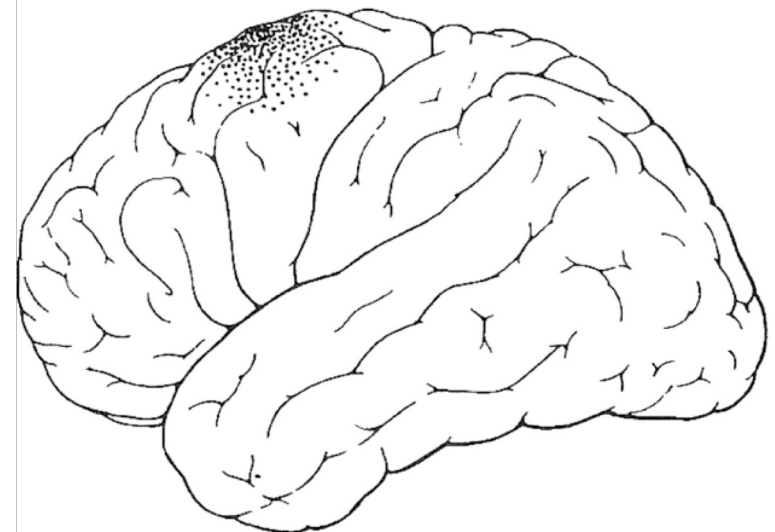
Supplementary Motor Area and Other Cortical Areas in Organization of Voluntary Movements in Man

P. E. ROLAND, B. LARSEN, N. A. LASSEN, AND E. SKINHOJ

Finger movement sequence (performance)



Finger movement sequence (mental rehearsal)



Movement planning

Motor Cognition

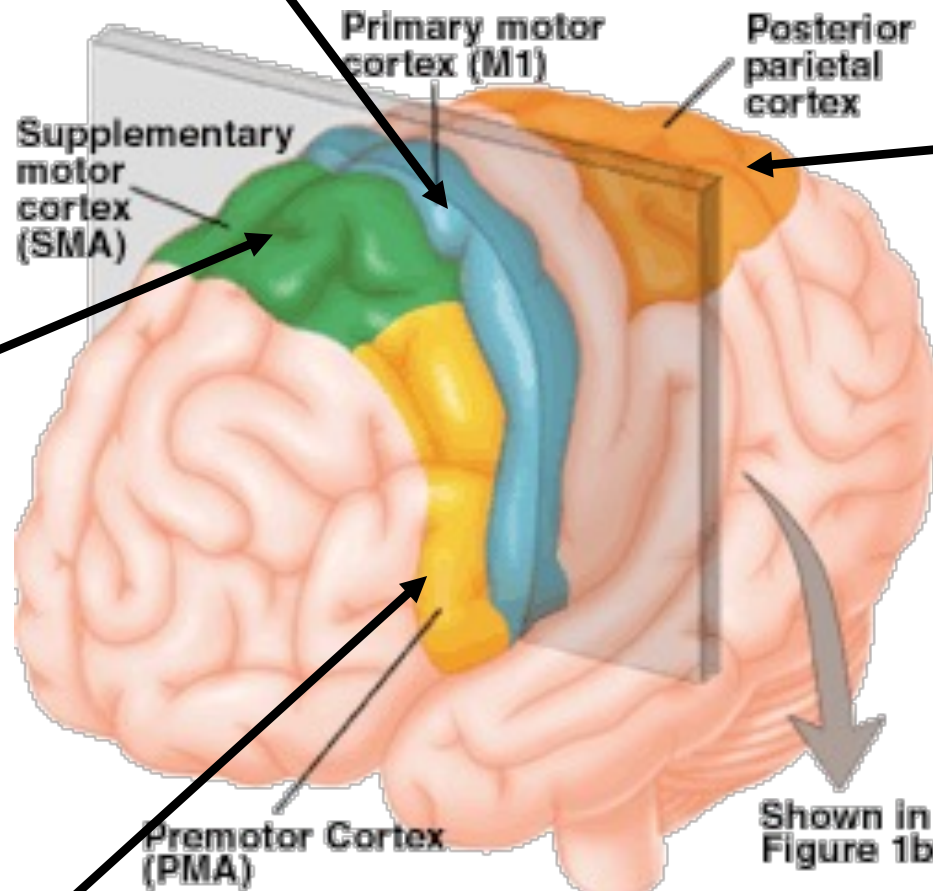
Execution

~ 50 ms

**Spatial
Information**

Planning

~1 sec

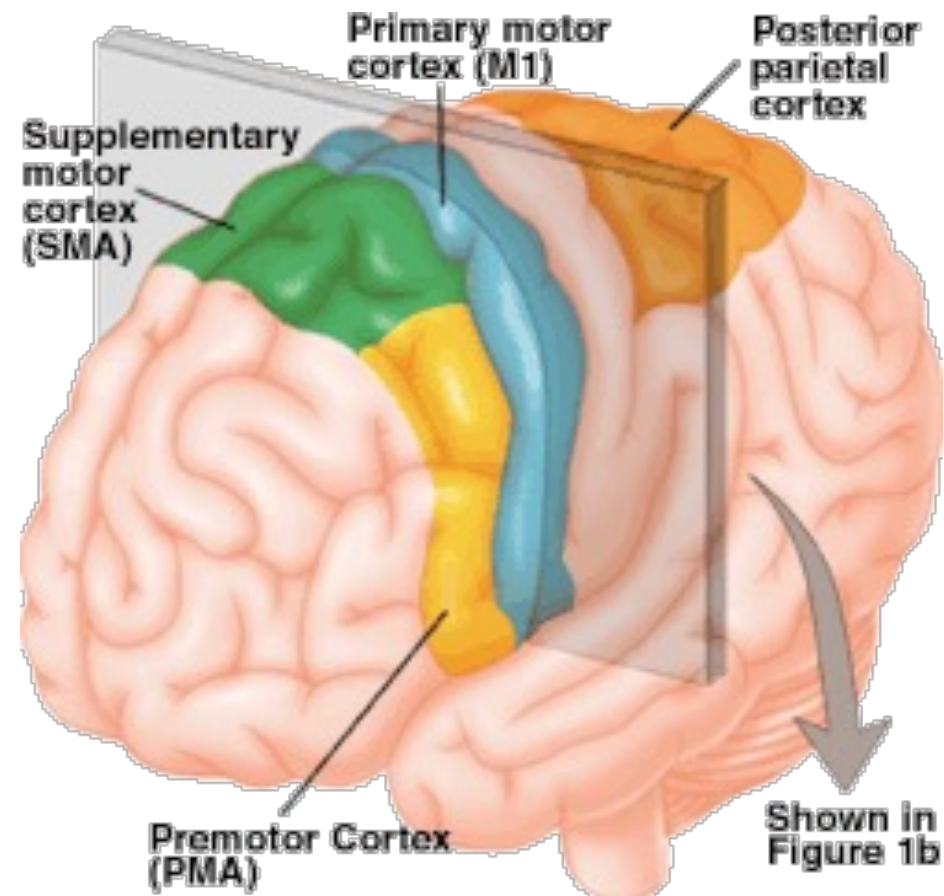


<https://brainconnection.brainhq.com/2013/03/05/the-anatomy-of-movement/>

Preparation

~120-150 ms

Parietal Cortex



<https://brainconnection.brainhq.com/2013/03/05/the-anatomy-of-movement/>

Apraxia



<https://www.thespruceeats.com/history-and-types-of-british-tea-435180>

Apraxia

How to prepare the perfect cup of tea?

- **Ideational**
- **Ideomotor**

Spatially relevant behavioral intentions